



Dolphin Technology

LTspice

Dolphin

Inverter

Revision 1.0

6 September 2023



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Table 1: Inverter gate truth table

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REVISION HISTORY

Revision	Date	Description of Changes
1.0	06-08-2023	



1. Inverter Gate

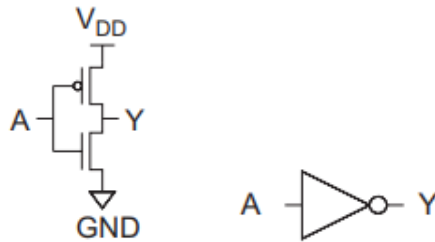


Table 1: Inverter gate truth table

A	Y
0	1
1	0

The inverter gate use one Nmos transistor and one Pmos transistor.

When input $A = 0$, Nmos transistor OFF and Pmos transistor ON, output $Y = 1$ (Y connect to VDD by Pmos).

When input $A = 1$, Nmos transistor ON and Pmos transistor OFF, output $Y = 0$ (Y connect to GND by Nmos).



2. Getting started using LTspice for simulation inverter gate

Download LTspice from website in the link below.

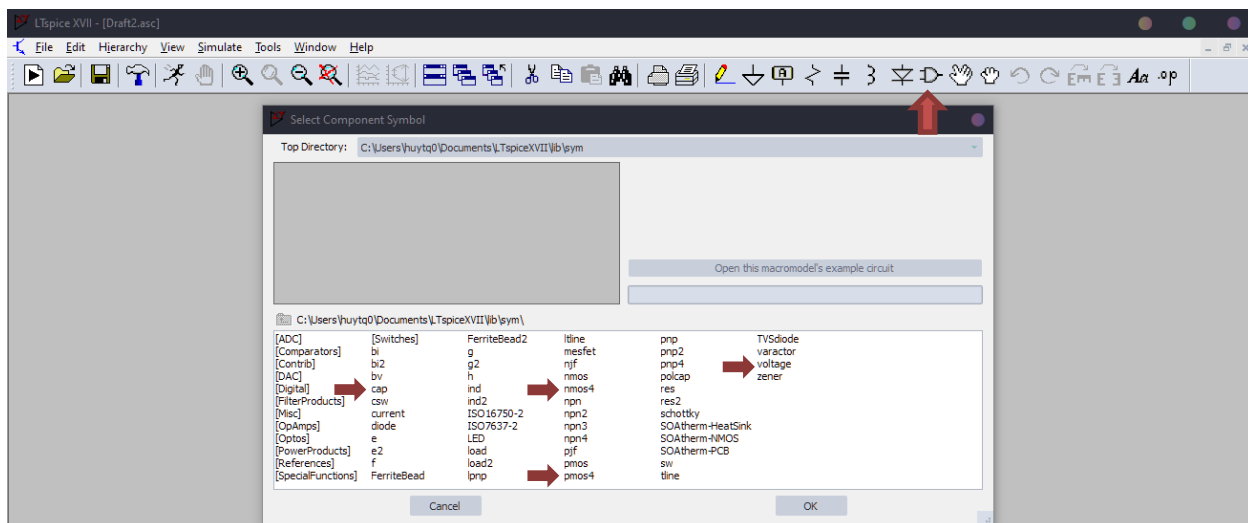
<https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html>

2.1. Schematic Circuit

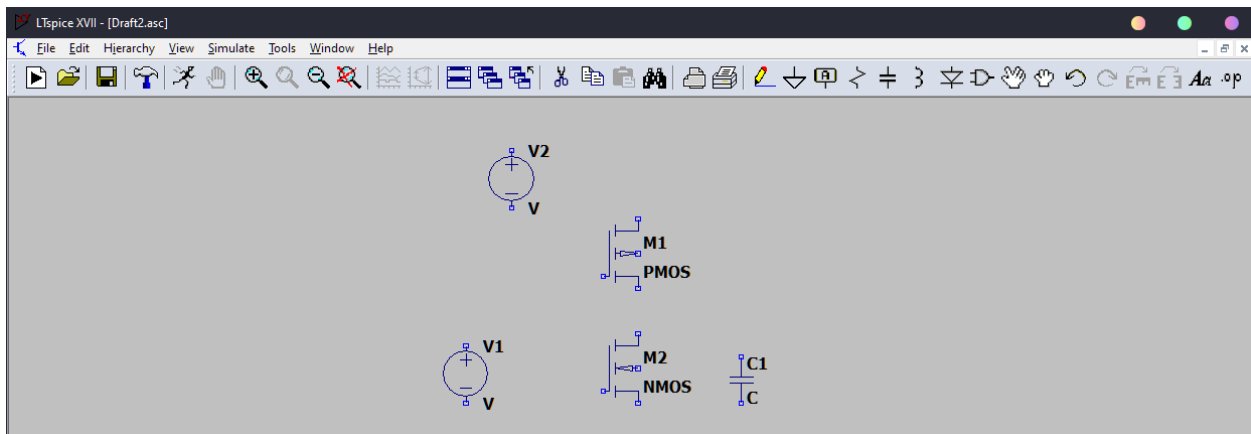
- Click on the “**New Schematic**” symbol to create new panel.



- Click on the “**Component**” symbol to show Component Panel and pick “**cap, nmos4, pmos4, voltage**”.

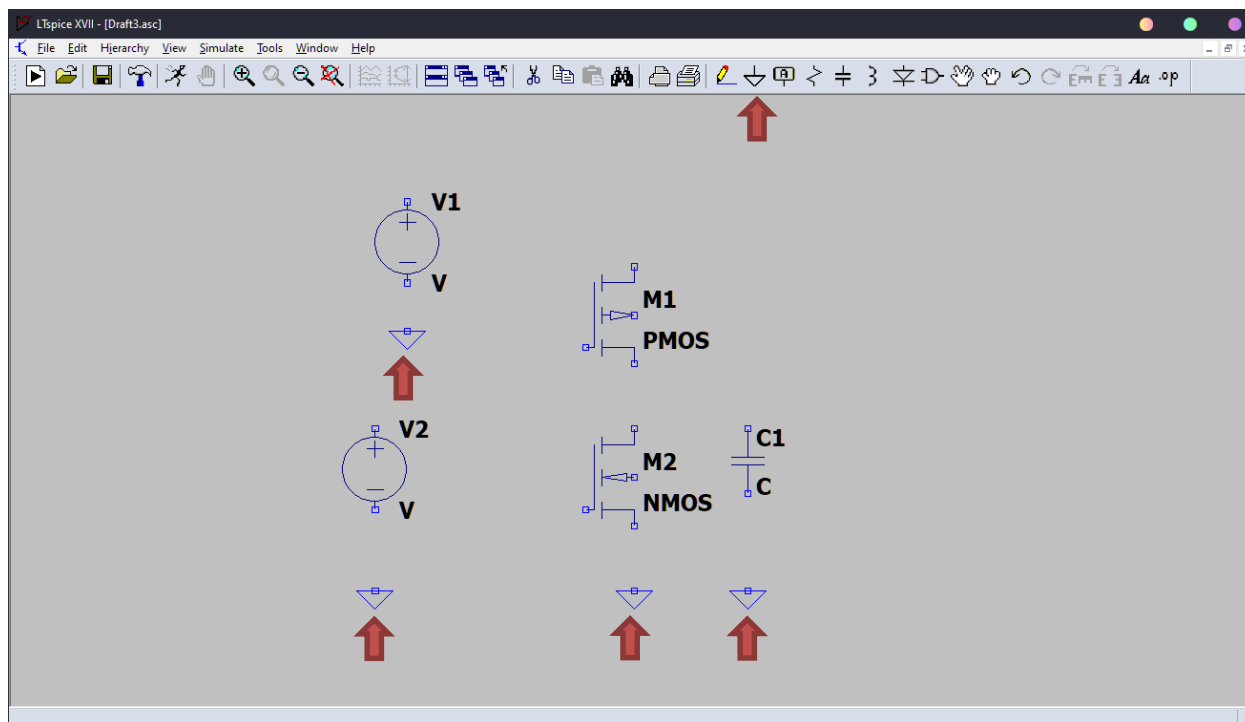


- The circuit need 1 pmos, 1 nmos, 1 cap and 2 voltages.

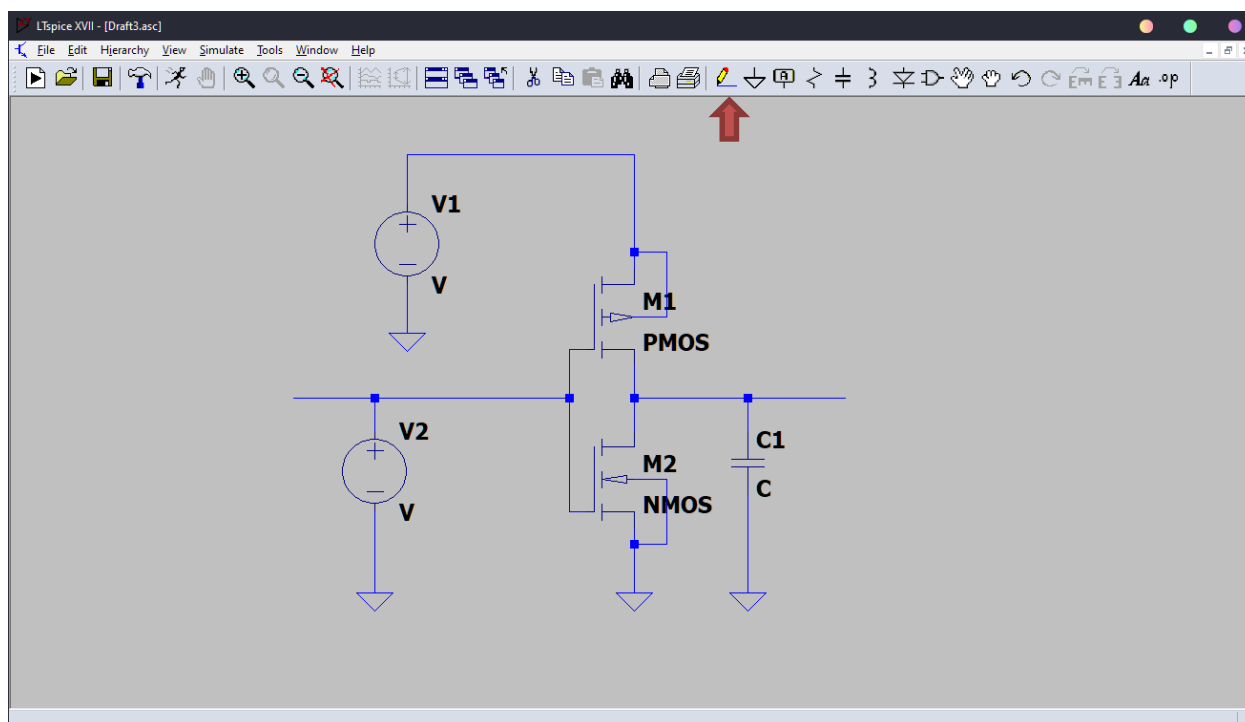




➤ Click on the “**Ground**” symbol and place 4 grounds.

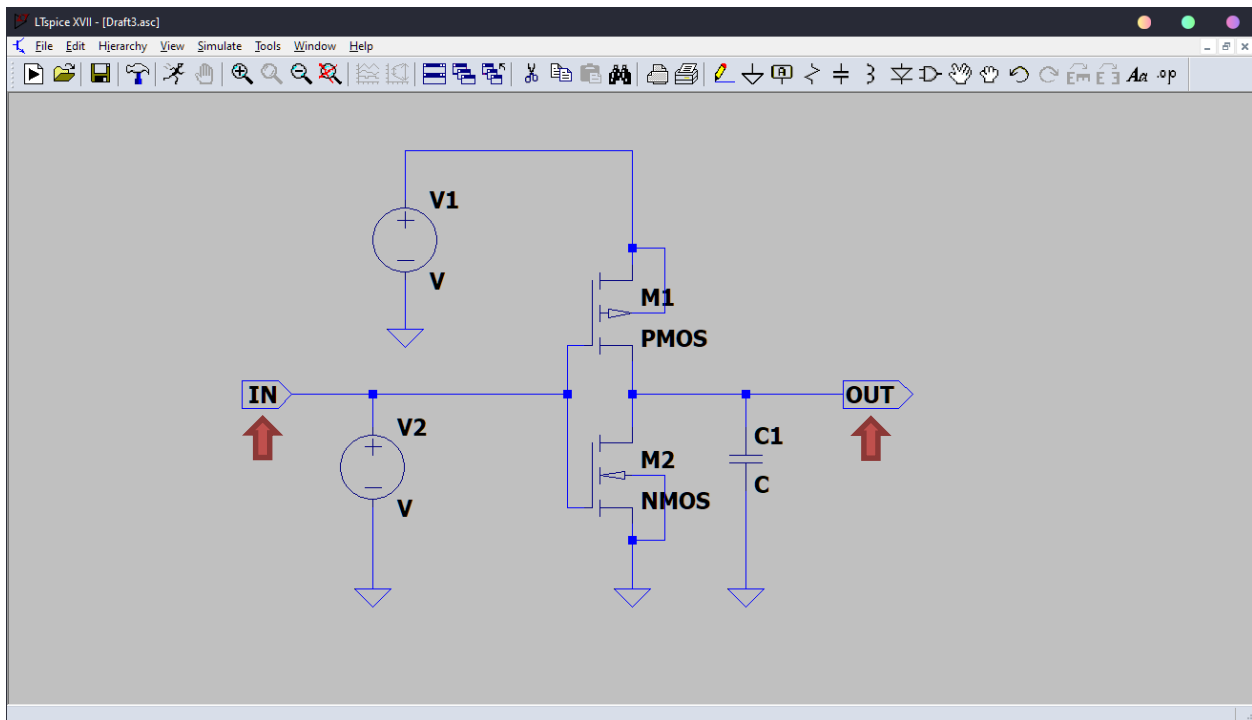
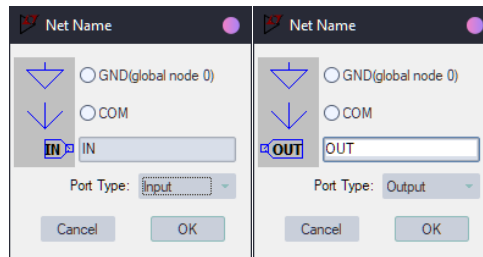


➤ Click on the “**Wire**” symbol to draw all connects.





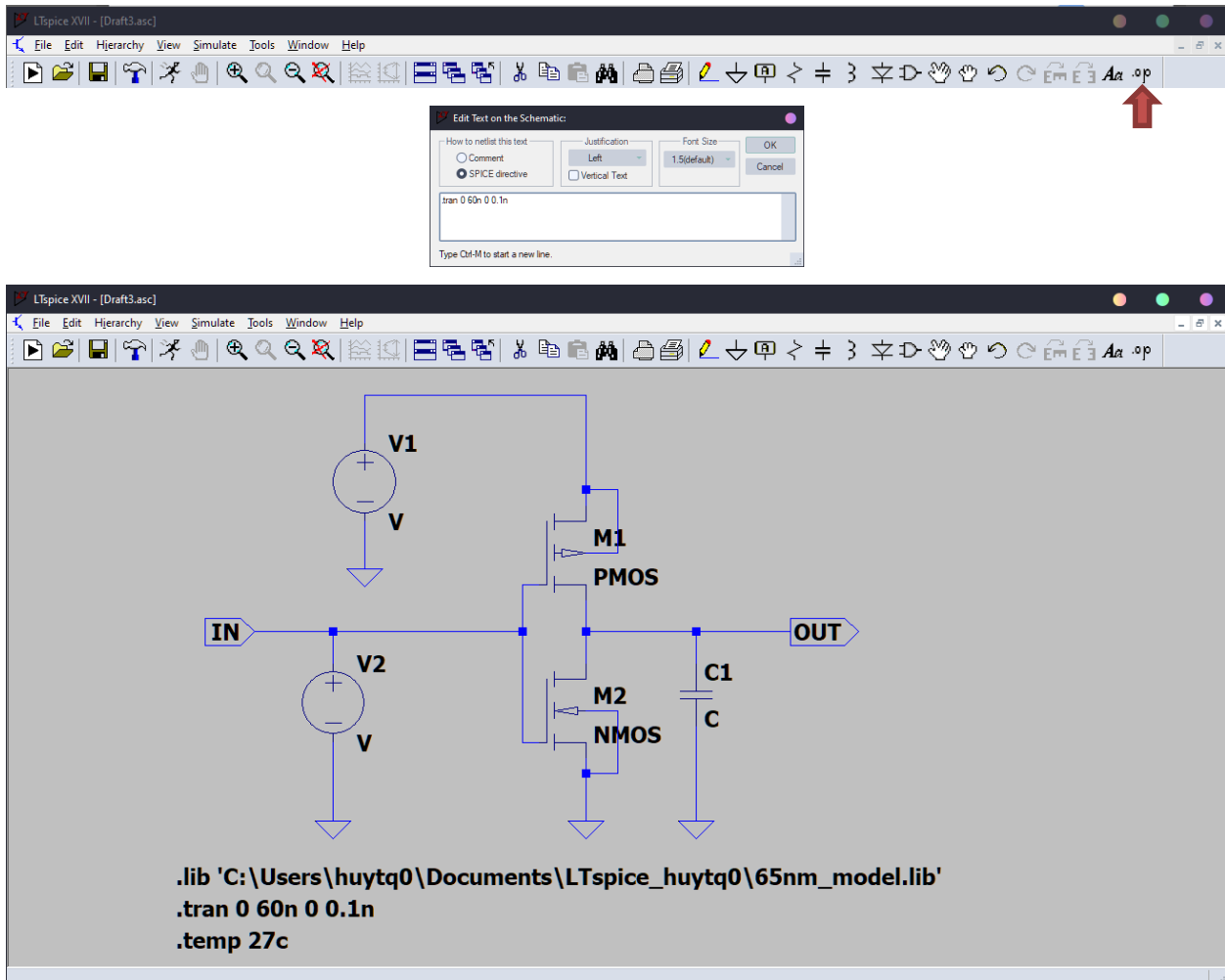
➤ Click on the “**Label Net**” symbol, choose type and name for Input and Output.





2.2. Simulation

- Click **“.op”** to add simulate option.
- Add transition time option, temperature option and library option.



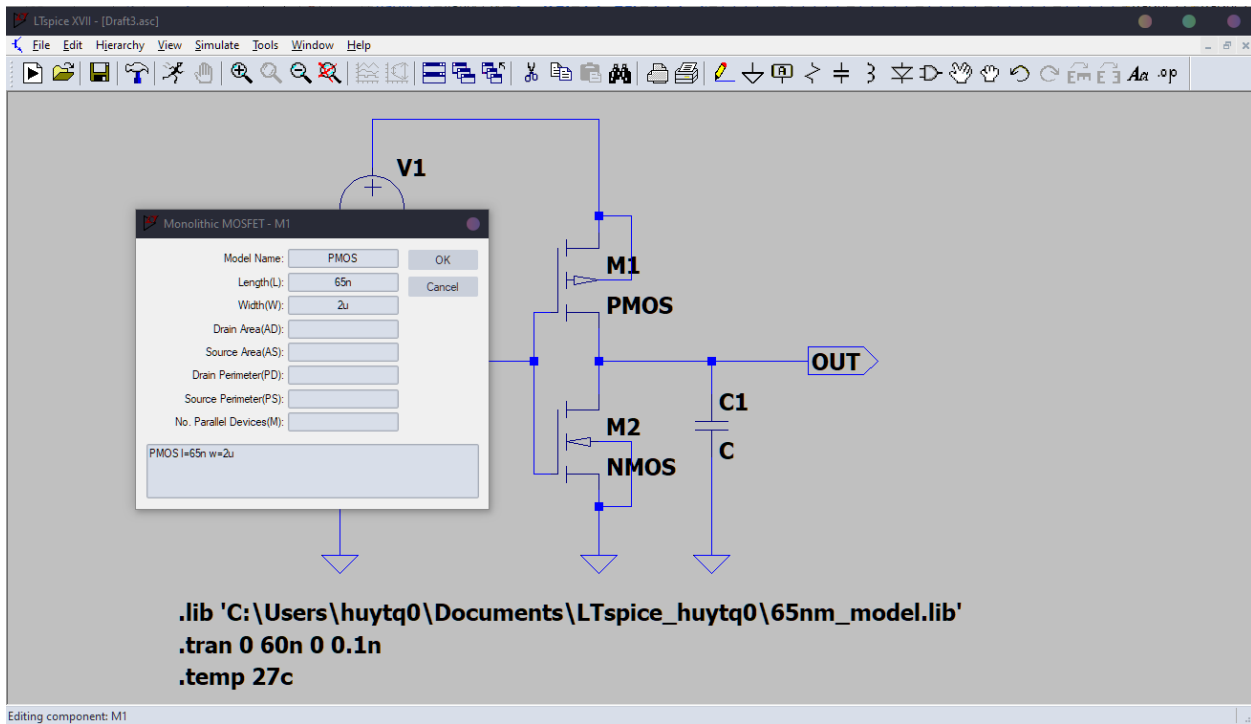
The text **“.tran 0 60n 0 1n”** mean simulation time is 60 nanosecond and time step is 0.1 nanosecond.

The text **“.temp 27c”** mean temperature is 27°C.

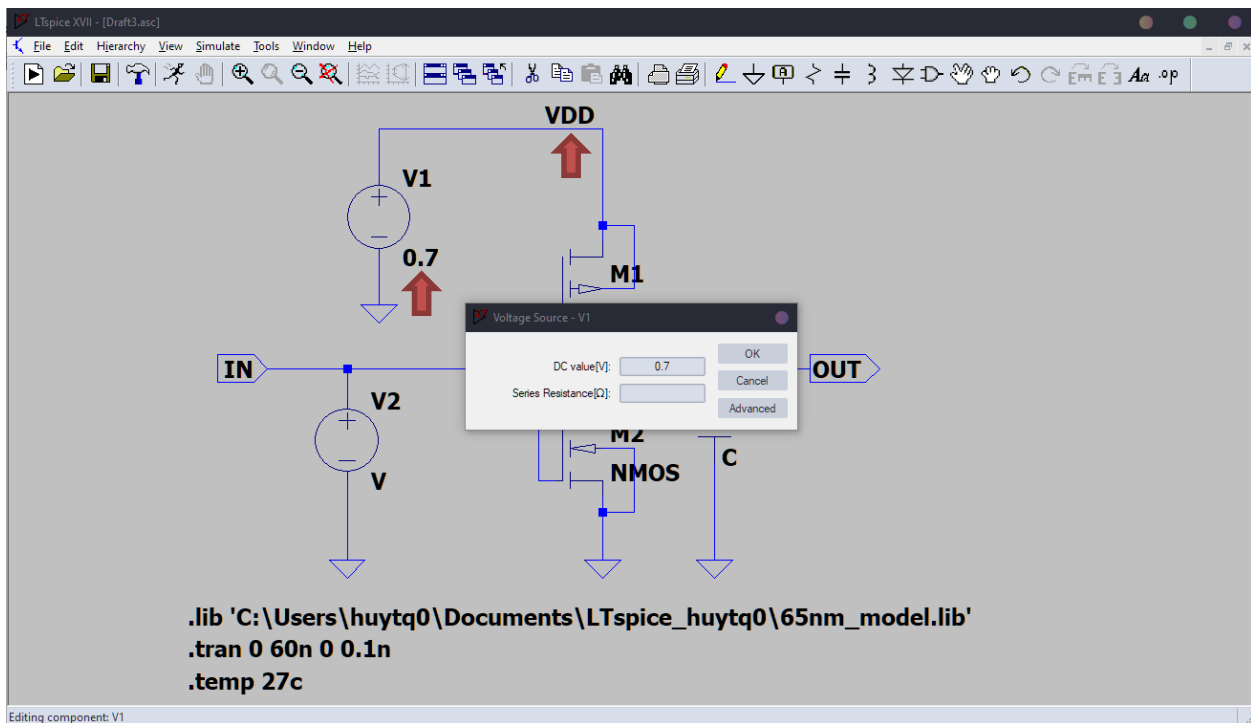
The text **“.lib ‘C: \Users\huytq0\Documents\LTspice_huytq0\65nm_model.lib’”** mean Cmos transistor technology model used in this simulation is 65 nanometer. Please add full path of 65nm_model.lib file.



- Right click on M1 PMOS. Edit Length is 65 nanometer and Width is 2 micrometer. Then edit M2 NMOS with the same value.

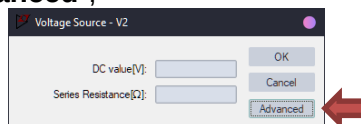


- Right click on V1. Edit DC value is 0.7 voltages and add a label name VDD.

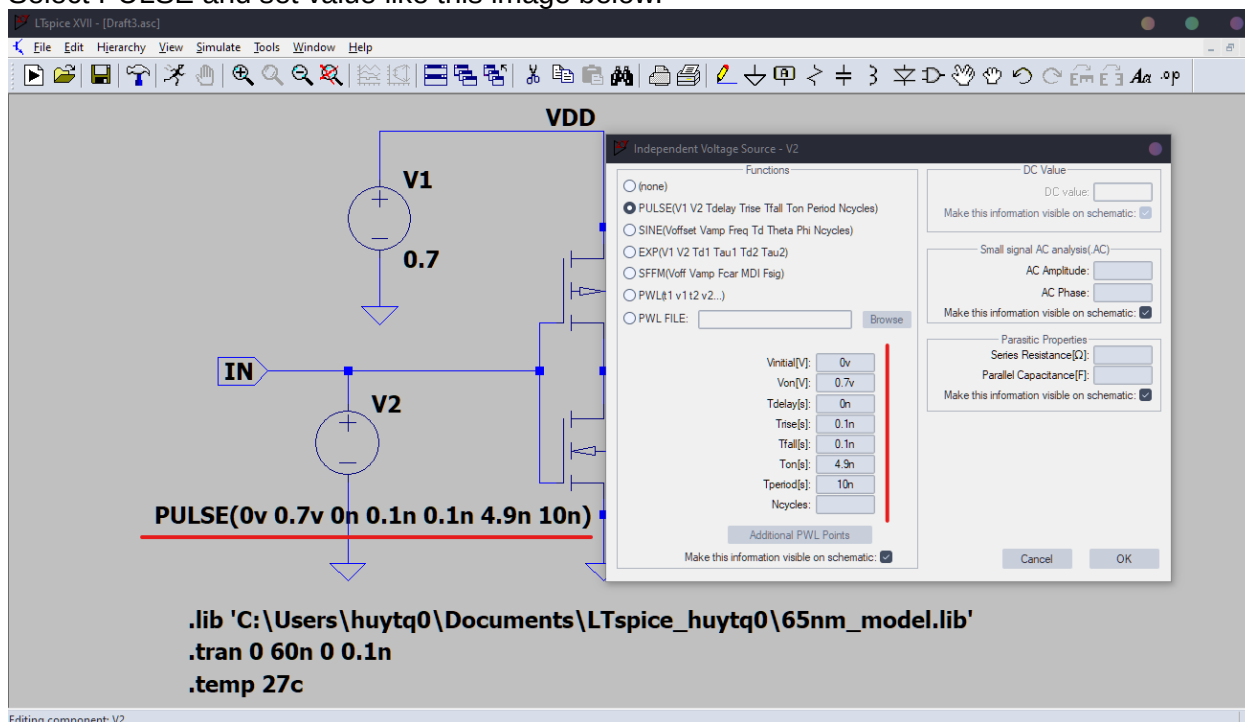




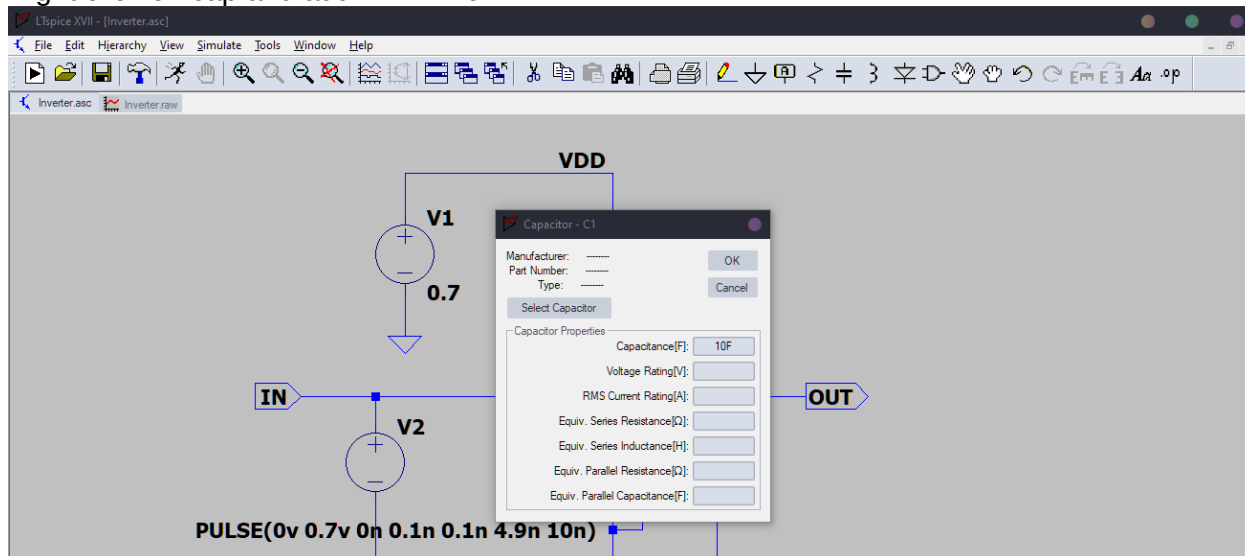
- Right click on V2 and choose “Advanced”,



- Select PULSE and set value like this image below.

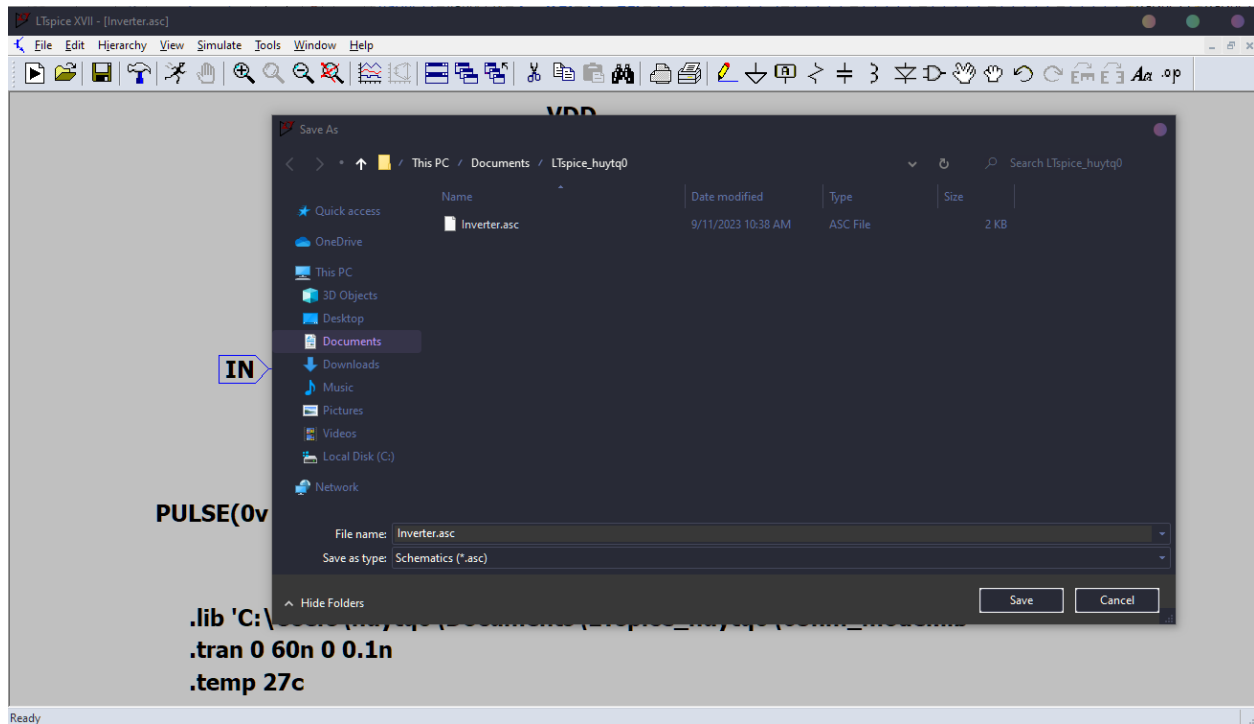


- Right click on cap and add value “10F”



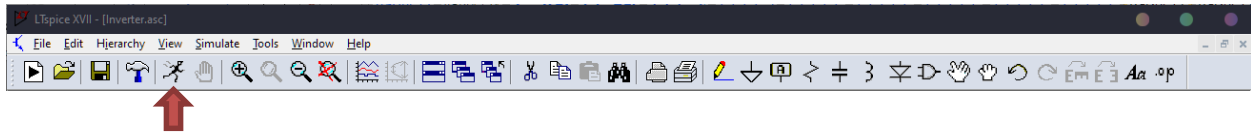


- After all, save the file click File -> Save as.

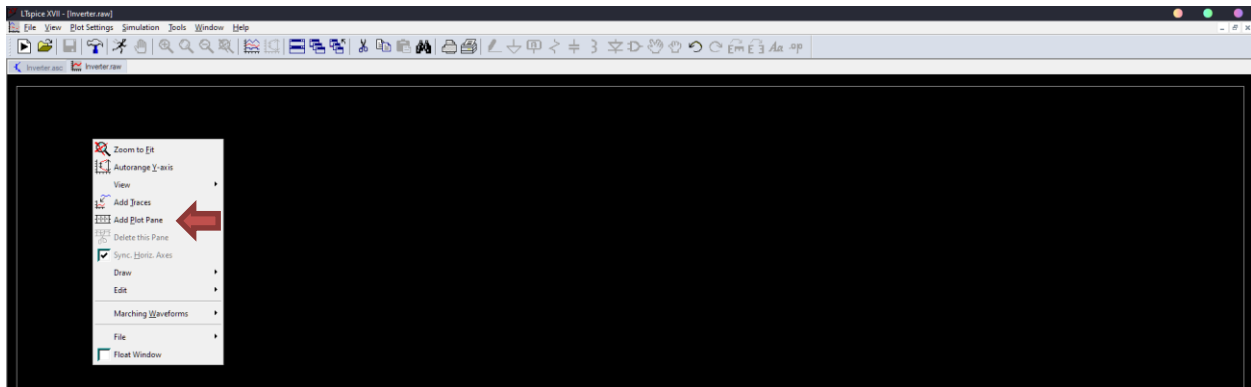




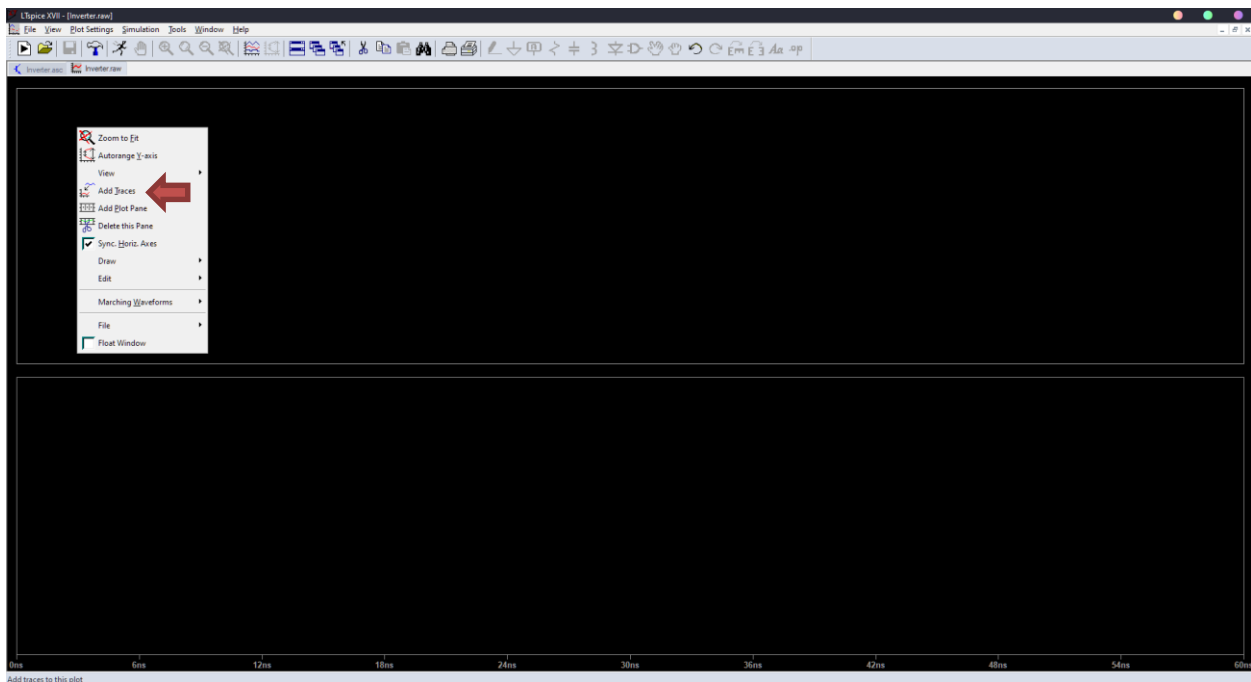
- Click on “**Run**” symbol to run simulation. After run simulation LTspice show a wave form window.



- In wave form window right click and select “**Add Plot Pane**” to add another pane.

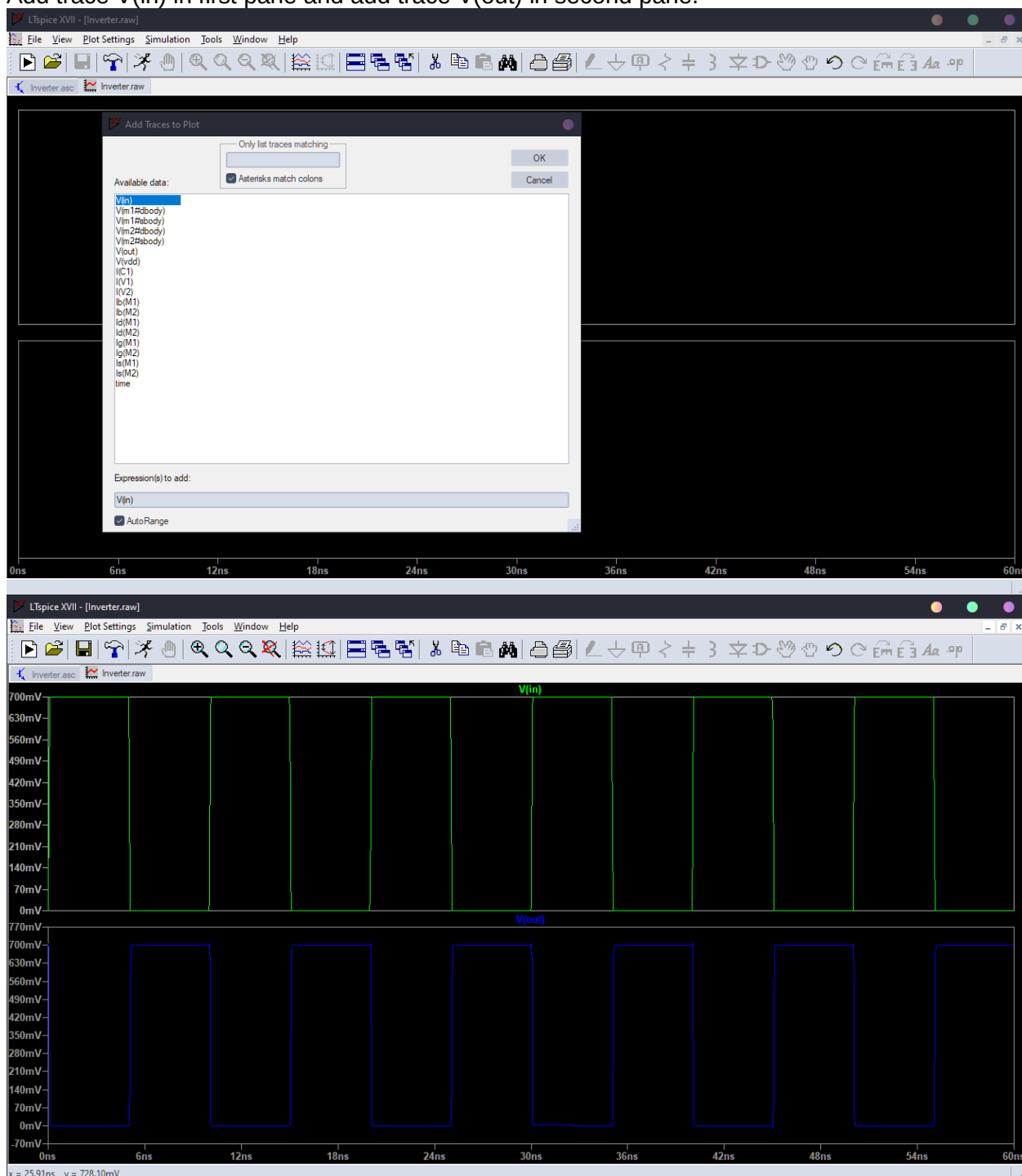


- Right click again and select “**Add Traces**”.





- Add trace V(in) in first pane and add trace V(out) in second pane.





2.3. Function and Measurement

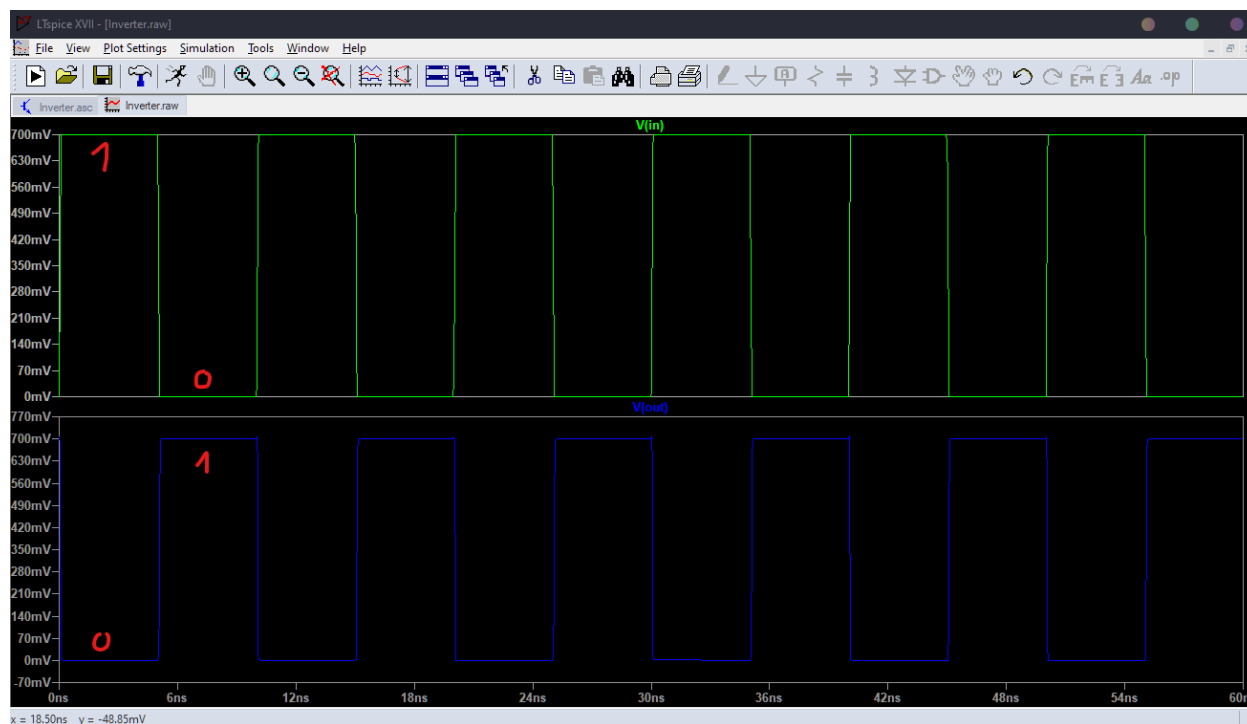
➤ Function

Let see in the wave form.

When $V(in) = 1$, $V(out) = 0$.

When $V(in) = 0$, $V(out) = 1$.

Inverter gate behavior is true.



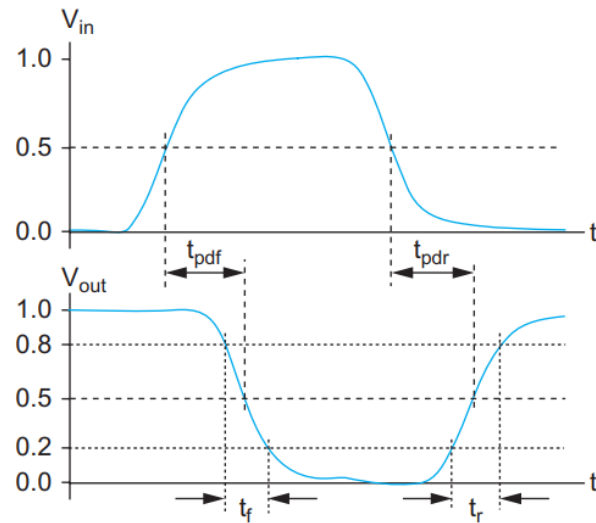


➤ Measurement

Propagation delay time, t_{pd} = maximum time from the input crossing 50% to the output crossing 50%.

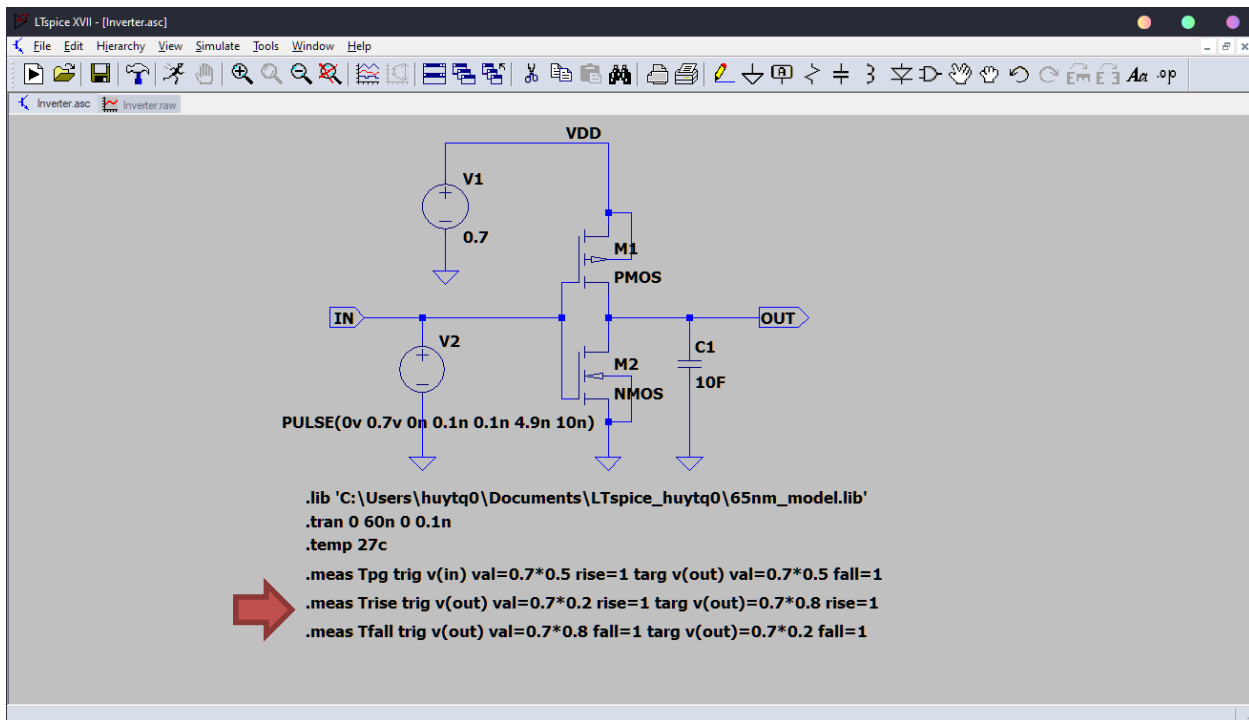
Rise time, t_r = time the output rise from 20% to 80% of its steady-state value.

Fall time, t_f = time the output fall from 80% to 20% of its steady-state value.





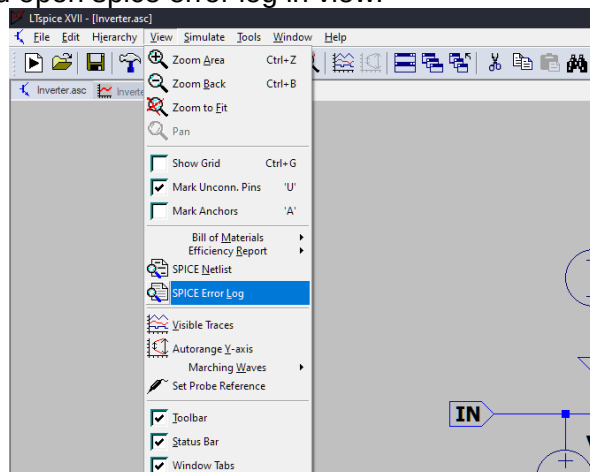
➤ In schematic window, add 3 measurement options.



The text “.meas Tpg trig v(in) val=0.7*0.5 rise=1 targ v(out) val=0.7*0.5 fall=1” mean: measure at the start point when v(in)=0.7*0.5 rise the first time and the end point when v(out)=0.7*0.5 fall the first time.



- Run simulation again and open spice error log in view.



- The result of measurement

```
SPICE Error Log: C:\Users\huytq0\Documents\LTspice_huytq0\Inverter.log
Circuit: * C:\Users\huytq0\Documents\LTspice_huytq0\Inverter.asc
Direct Newton iteration for .op point succeeded.
tpg=2.39315e-011 FROM 5e-011 TO 7.39315e-011
trise=3.40143e-011 FROM 5.07345e-009 TO 5.10746e-009
tfall=2.43845e-011 FROM 6.0469e-011 TO 8.48535e-011

Date: Mon Sep 11 15:13:20 2023
Total elapsed time: 0.070 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 2038
traniter = 2034
tranpoints = 909
accept = 829
rejected = 80
matrix size = 13
fillins = 0
solver = Normal
Matrix Compiler1: 960 bytes object code size 0.2/0.1/[0.1]
Matrix Compiler2: 1.32 KB object code size 0.1/0.2/[0.1]
```